AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I107*

Team Name: *QuadraMind*

Domain: *#7 Artificial Intelligence and Cybersecurity*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Real-Time Motor Fault Detection and Cyberattack Identification

Sensor Reliability

Traditional systems may **trust sensor data** without *verifying it*.

Motor Faults

Bearing damage and **mechanical wear** can alter *signal patterns*.

Cyberattacks

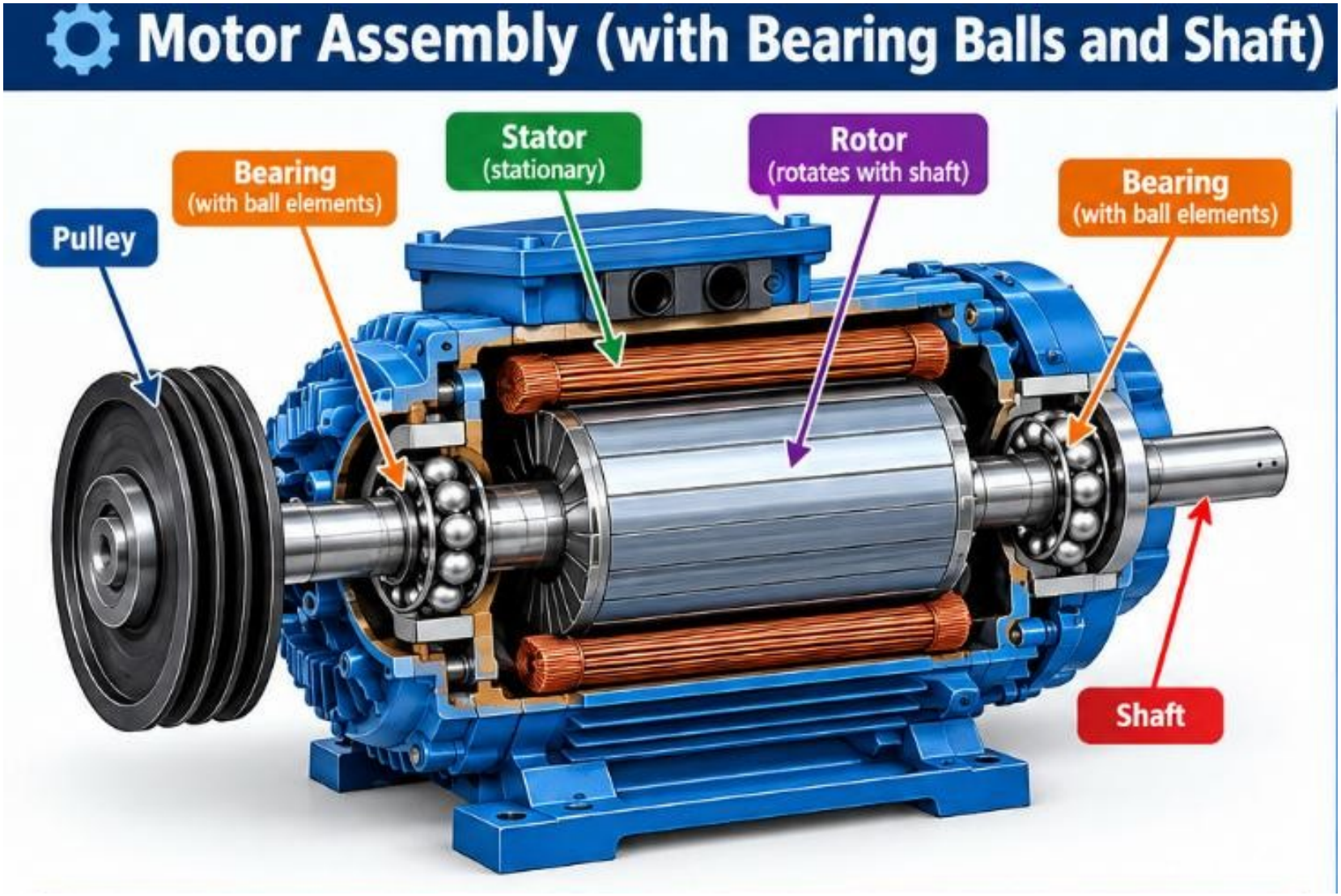
False-data injection and **replay attacks** can *manipulate readings*.

Signal Confusion

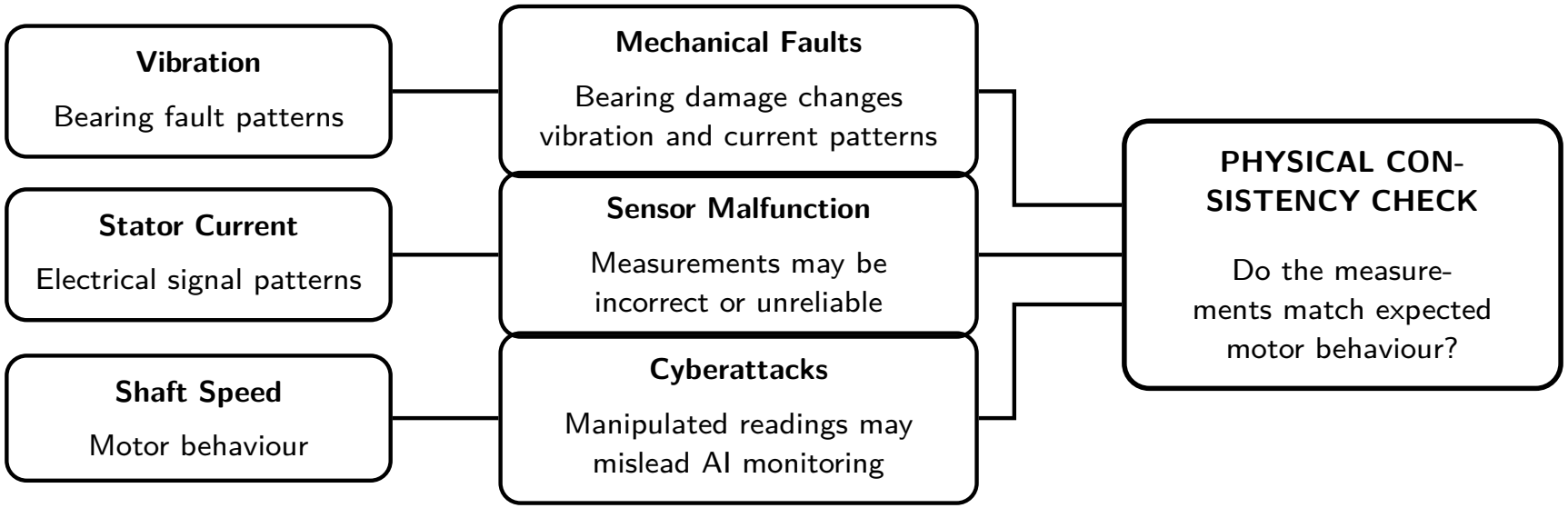
Faults, **sensor failures**, and **attacks** may produce *similar patterns*.

Our Goal:

Develop a **smart system** that distinguishes **genuine motor faults**, **sensor malfunctions**, and **suspected cyberattacks** using *existing motor signals*.


Motor Assembly (with Bearing Balls and Shaft)
Pulley
Bearing
(with ball elements)
Stator
(stationary)
Rotor
(rotates with shaft)
Bearing
(with ball elements)
Shaft

Understanding Motor Signals and Their Risks



Key Idea:

Combining multiple motor signals with physical consistency checking helps identify faults, unreliable sensors, and suspicious data.

Existing Gaps & Challenges

- Conventional systems classify conditions as normal or abnormal without identifying the actual cause.
- Bearing faults, sensor failures, and cyberattacks can produce similar signal patterns.
- Existing approaches often assume sensor data is trustworthy.
- Additional reference sensors increase hardware cost and system complexity.
- Real-time detection on low-cost edge devices is challenging.
- A unified solution for fault diagnosis, sensor validation, and cyberattack detection is needed.

Real-Time Motor Fault Detection Using Dynamic Watermarking and AI

Trust-Aware Monitoring

Develop a trust-aware motor monitoring system to distinguish genuine motor faults from sensor failures and cyberattacks.

Multi-Sensor Cross-Check

Cross-check vibration, stator current, and shaft speed using physical consistency relationships.

Dynamic Watermarking

Incorporate Dynamic Watermarking by introducing a small, controlled, known signal variation into a suitable motor control input and observing the response to help verify system integrity.

Signal Feature Extraction

Apply FFT, Hilbert Transform, Kurtosis, and Envelope Analysis to extract meaningful signal features.

Attack Detection

Detect False Data Injection and Replay Attacks by identifying inconsistencies between expected and observed motor responses.

Machine Learning / CNN-based classification

Use Machine Learning / CNN-based classification to identify motor conditions and suspicious data manipulation.

Generate four outputs: Healthy | Bearing Fault | Sensor Fault | Suspected Cyberattack.

Real-Time Edge Deployment

Target real-time edge deployment using existing motor signals, with no additional physical sensors where feasible.

Integrated Motor Fault Detection Using Multi-Sensor Fusion, Dynamic Watermarking, and AI

No.	Technical Stage	Description
1	**Multi-Sensor Data Acquisition**	Collect vibration, stator current, and shaft-speed signals.
2	**Signal Preprocessing**	Apply noise reduction, filtering, and signal normalization.
3	**Feature Extraction**	Use FFT, Hilbert Transform, Kurtosis, and Envelope Analysis to extract fault-related features.
4	**Multi-Sensor Fusion**	Combine sensor data and cross-check physical relationships to identify inconsistencies.
5	**Dynamic Watermarking**	Introduce a small, controlled variation into a suitable motor control input and compare the observed response with the expected watermark signature.
6	**AI-Based Classification**	Use a CNN or lightweight machine learning model to classify extracted signal patterns. Integrate fault features and data-trust indicators to distinguish genuine faults from unreliable sensor data or suspected attacks.
7	**Output and Alert Generation**	Classify the motor condition into four states: Healthy, Bearing Fault, Sensor Fault, and Suspected Cyberattack. Generate maintenance alerts or cybersecurity investigation requests.
8	**Deployment**	Target real-time operation on a low-cost microcontroller or edge device.

Innovation & Uniqueness

- **Multi-Sensor Fusion:** Combines vibration, stator current, and shaft speed for comprehensive motor condition assessment.
- **Dynamic Watermarking:** Uses an active challenge-and-response mechanism to help verify sensor data integrity and investigate cyberattacks.
- **Trust-Aware AI:** Integrates motor fault classification with sensor-data trust assessment.
- **Integrated Diagnosis:** Identifies four conditions: Healthy, Bearing Fault, Sensor Fault, and Suspected Cyberattack.
- **Improved Reliability:** Aims to reduce false alarms and distinguish genuine mechanical faults from unreliable measurements or suspicious data manipulation.
- **Edge-Based Deployment:** Targets real-time monitoring on low-cost hardware, without additional physical sensors where feasible.
- **Cause-Specific Decisions:** Supports appropriate actions such as motor repair, sensor replacement, or cybersecurity investigation.

Feasibility & Scalability

Category	Description
Technical Feasibility	■ Uses vibration sensors and existing motor signals for data acquisition. ■ Applies established signal-processing techniques: FFT, Hilbert Transform, Kurtosis, and Envelope Analysis. ■ Uses a Convolutional Neural Network (CNN) for automated fault classification. ■ Python or MATLAB can be used for development, testing, and validation. ■ Lightweight models can potentially be deployed on low-cost edge devices.
Scalability	■ Can be adapted to different motor types after suitable testing and model training. ■ Can be extended to industrial machinery, pumps, and wind turbines. ■ Can support continuous motor condition monitoring. ■ Future IoT integration can enable remote monitoring.

Cost for Product (Optional)

Estimated Prototype Cost

Component	Estimated Cost ()
Vibration sensor	500–1,500
Microcontroller / Edge board	500–2,000
Power supply and connecting components	300–700
Mounting, wiring, and accessories	300–800
Estimated hardware total	**1,600–5,000**

Additional Considerations

- Python and many signal-processing and machine-learning tools are free.
- Existing computers can be used for initial model development.
- Actual costs depend on sensor quality, hardware selection, and prototype requirements.
- Industrial deployment may require additional hardware, enclosure protection, calibration, and testing.

Cost Disclaimer

Preliminary planning estimates, not supplier quotations. Excludes labour, taxes, and existing motor or test-rig costs.

Use Cases

Use Case	Description
1. Manufacturing Industries	■ Detects potential motor faults before major equipment failure. ■ Helps reduce unexpected production interruptions.
2. Pumps and Water Treatment Plants	■ Monitors motor vibration to identify developing mechanical problems. ■ Supports reliable operation of pumping equipment.
3. Wind Turbines	■ Supports condition monitoring of motor-driven auxiliary equipment. ■ Helps maintenance teams identify abnormal vibration patterns.

Use Cases (Continued)

Use Case	Description
4. Industrial Maintenance	■ Identifies potential cracking faults, offset pulley conditions, and wear. ■ Helps technicians plan inspections and repairs.
5. Remote Motor Monitoring	■ Can be integrated with edge devices for local analysis. ■ Future IoT integration can support remote alerts and condition monitoring.

Impacts and Benefits

- **Technical Benefits**
  - Automated motor fault classification using CNN.
  - Improved vibration analysis using multiple signal-processing techniques.
  - Potential for early identification of developing mechanical faults.
- **Economic Benefits**
  - Helps reduce unexpected equipment downtime.
  - Supports maintenance planning and efficient resource use.
  - May reduce repair costs through earlier fault identification.
- **Operational Benefits**
  - Reduces dependence on manual vibration analysis.
  - Provides consistent, data-driven diagnostic support.
  - Helps prioritise inspections and repairs.
- **Safety and Reliability**
  - Helps identify abnormal machine conditions before they escalate.
  - Supports safer and more reliable operation with appropriate maintenance procedures.

Expected Impact

Improved motor condition monitoring, informed maintenance decisions, and reduced operational disruption.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	9.4	Industry, Innovation and Infrastructure	Promotes intelligent industrial monitoring, AI-based fault diagnosis, and predictive maintenance.
SDG 12	12.5	Responsible Consumption and Production	Supports timely maintenance, equipment longevity, and efficient use of machinery and replacement parts.
SDG 7	7.3	Affordable and Clean Energy	Supports reliable motor-driven equipment and may help reduce energy waste from poorly maintained machinery.
SDG 8	8.2	Decent Work and Economic Growth	Supports industrial productivity, efficient maintenance, and reduced unplanned production

Individual Member Contribution

Team Member	Contribution
S. Sai Swathee	Research on motor faults, study of dynamic watermarking and identification of project requirements, finding methodology or working principle for the solutions, PPT making.
V. Harshini	Signal-processing techniques for motor fault detection, finding the research papers and gaps in existing papers, PPT making.
Sridha Anbukumar	Study of CNN-based fault classification, checking novelty and study of sensor data acquisition, PPT making.
Niveditha Krishna	System validation, presentation preparation, finding the research papers and cyberattack detection, and maintaining enthusiasm.

HACKOTHAN PICTURES



Yukti ID

IR2026-1051079